Sahaj Bhadja

(213) 284-4651 bhadja@usc.edu sahaj-bhadja sahajbhadja sahaj-bhadja

Summary

Software Engineer with 2+ years building **reliable system software** and **test-driven backend services** using **C++**, Python, SQL, and cloud tooling, with focus on **debugging**, **integration**, and production-quality delivery.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Software Engineering, Operating Systems, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 - Present*
- Built and maintained backend data workflows across **Azure SQL/PostgreSQL** for 500+ users, reducing latency from **1.2s to 350ms** through query tuning and schema improvements.
- Implemented validation/release checks for 5K+ monthly jobs with reconciliation logic, improving **reliability**, defect prevention, and support readiness.

Grad8 **LA, California**
Software Engineer Intern *June 2025 - August 2025*
- Developed and tested Flask + React features handling **10K+ daily requests**, translating functional requirements into modular APIs and secure workflows.
- Added concurrent processing and CI/CD controls via GitHub Actions, reducing turnaround time by **60%** while maintaining stable peak-load behavior.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 - May 2025*
- Built Python experimentation/simulation pipelines processing **100K+ trajectory logs** with reproducible test runs and performance tracking.
- Produced technical documentation for validation results and system behavior, improving traceability and cross-team issue resolution.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 - July 2024*
- Developed and maintained 20+ unit/integration test suites using Jenkins, PyTest, and Mockito across 3 OS environments to strengthen SDLC quality gates.
- Troubleshot API/ETL defects using OOP, AWS Glue, and S3, improving response times by **30%**, boosting pipeline efficiency by **68%**, and reducing technical debt by **20%**.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 - August 2023*
- Built MERN application modules with REST APIs and WebSockets supporting 8+ workflows and 2M+ synchronized records, including clear operational logging.
- Automated deployment/testing with Jenkins and Docker, reducing deployment time by **50%** and improving release consistency during change cycles.

Skills

Languages: **C++**, Python, Java, JavaScript/TypeScript, SQL
Embedded/System Foundations: OOP, operating systems, multithreading basics, debugging, system integration, validation workflows
Backend & APIs: Flask, FastAPI, Node.js, Express.js, REST APIs, JSON
Databases: PostgreSQL, MySQL, SQL Server, MongoDB, Azure SQL
Tooling: Git, GitHub Actions, Jenkins, Docker, Kubernetes, CI/CD, unit/integration testing
Engineering Practices: SDLC, Agile/Scrum, test design, defect triage, technical documentation

Projects

CPU-Scheduling Simulator Live | *C++, React.js, JavaScript*
- Implemented 6 scheduling algorithms with interactive controls and live visualizations across 1K+ simulated process combinations, improving systems debugging and verification.

Chat2SQL Code | *Java, TypeScript, SQL, Maven, Docker*
- Built an AI-powered SQL workspace supporting 10+ databases, accelerating query authoring, schema exploration, and structured execution validation workflows.

HMIS Live | *MERN, ABDM APIs, Twilio*
- Designed structured healthcare workflows with API integrations and monitoring-backed operations supporting 100+ daily appointments and improved record retrieval by 2x.